MIL-STD-294A
15 July 1969

SUPERSEDING
MIL-STD-294
2 November 1956

MILITARY STANDARD

VISUAL INSPECTION GUIDE FOR RUBBER V BELTS



FSC 3030

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

MIL-STD-294A
15 July 1969

DEPARTMENT OF DEFENSE

WASHINGTON, D. C. 20301

Visual Inspection Guide for Rubber V-Belts
MIL-STD-294A

1. This Military Standard is mandatory for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions, or deletions, should be addressed to the Naval Ship Engineering Center, Center Building, Prince George's Center, Hyattsville, Maryland 20782.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

FORWARD

Government material procurement specifications and standards are primarily concerned with detailing composition, construction and necessary requirements to satisfy the demand for dependability and longevity. The visible defects shown in this document are guides for inspectors to evaluate quality and workmanship of the finished product. This Military Standard is not intended to take the place of requirements, tests and examinations specified in V-belt procurement specifications and documents. The quality requirements specified in the basic specification are governing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

CONTENTS

			Page
Paragraph	1	SCOPE	1
	1.1	Scope	1
	1.2	Purpose	1
	2	REFERENCED DOCUMENTS	1
	3	DEFINITIONS	1
	3.1	Major and minor defects	1
	3.2	Work descriptions	1
	3.3	Angle	1
	3.4	Cords	1
	3.5	Cogged or notched	1
	3.6	Cover	1
	3.7	Compression member	1
	3.8	Indentation	1
	3.9	Longitudinal seam	1
	3.10	Mandrel	1
	3.11	Matched belts	1
	3.12	Misidentification	2
	3.13	Tension member	2
	3.14	Transverse seam	2
	4	GENERAL REQUIREMENTS	3
	4.1	Sampling	3

FIGURES

Figure	1.	Indentation on bottom of belt -- major	4
	2.	Indentation on top of belt -- major	5
	3.	Indentation on top of belt -- major	6
	4.	Indentation on top of belt -- major	7
	5.	Indentation on top of belt -- major	8
	6.	Light spot -- major	9
	7.	Light spot -- minor (upper limit for minor, any larger becomes major)	10
	8.	Press lap defect -- major	11
	9	Press lap defect -- major	12
	10.	Flow distortion -- major	13
	11.	Flash distortion -- major	14
	12.	Flash untrimmed -- major	15
	13.	Misplaced brand with twist -- major	16
	14.	Misplaced brand without twist -- minor (Left belt has misplaced brand. Right belt is a correct one for comparison)	17
	15.	Protrusion -- major	18
	16.	Blister, any side -- major	19
	17.	Foreign material, top of belt -- minor	20
	18.	Foreign material, internal -- major	21
	19.	Hole in cogging area -- major	22
	20.	Porosity, between cord layer and upper part of belt -- major	23
	21.	Open longitudinal seam -- major	24
	22.	Open longitudinal seam, bottom of belt -- minor; if on wearing side -- major	25
	23.	Loose longitudinal seam -- major	26
	24.	Open transverse seam -- major	27
	25.	Open transverse seam -- major	28
	26.	Loose transverse seam -- major	29
	27.	Loose transverse seam, wearing side -- major, top or bottom -- minor	30
	28.	Torn cover -- major	31
	29.	Torn cover, bottom of belt -- minor; if on wearing side -- major	32
	30.	Trim cut in cover -- major	33

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

Page

Figure 31. Trim cut in cover -- minor. 34
32. Scratch, not through a ply -- minor 35
33. End scrap (Left belt has material missing due to being cut too near the end of the mandrel. Right belt is a correct one for comparison.) -- major. . . . 36
34. Wrinkle in cover -- minor 37
35. Cogging cut-outs skipped -- major 38
36. Cogging cut-out overlap -- minor. 39
37. Cogging missed -- major 40
38. Wavy wire in steel cable belt (X-ray view) -- major 41

TABLE

Table I - Classification of defects. . . . . 2 and 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard covers only visible defects for V-belts, that have a trapezoidal cross-section, manufactured with natural or synthetic rubber compounds and reinforced with cord, fabric or wire. Power is transmitted through the nonparallel sides of the belt. Timing belts or belts that transmit power through the inner face or bottom are excluded.

1.2 Purpose. The primary objective of this standard is to furnish descriptive words and photographs of deficiences that may occur during the manufacture of V-belts. This Standard is issued for the use of Federal and Military agencies to appraise, through visual examination, the acceptance or rejection of V-belts.

2. REFERENCED DOCUMENTS

2.1 The issues of the following documents in effect on date of invitation for bids form a part of this standard to the extent specified herein.

GOVERNMENTAL

MILITARY STANDARDS
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-177 - Rubber Products, Terms for Visible Defects of.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 Major and minor defects. Defects are defined as follows:

(a) Major. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.
(b) Minor. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

3.2 Word descriptions. Word descriptions of visible defects are in accordance with MIL-STD-177.

3.3 Angle. The angle formed at the intersection of the projections of the driving sides of the belt.

3.4 Cords. Twisted fibers of natural or synthetic material used to reinforce the belt structure.

3.5 Cogged or notched. A belt produced with a series of evenly spaced U- or V-shaped indentations in the inner circumference or bottom face.

3.6 Cover. The outer material of the belt, consisting of one or more plies of stretchable fabric or cord, impregnated with rubber compounds for adhesion, usually intended as a protective covering.

3.7 Compression member. The narrower trapezoidal portion of the belt in the section below the tension members that may contain cords or fabric.

3.8 Indentation. A recess on the belt surface.

3.9 Longitudinal seam. The junction point of material in the length of the finished belt.

3.10 Mandrel. A machined form to control size and to support the material during belt fabrication and vulcanization.

3.11 Matched belts. Two or more belts that have identical rotational lengths and cross-sectional areas within specified limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

3.12 Misidentification. Marking or identification incorrect, missing or illegible.

3.13 Tension member. Cord, fabric, or wire reinforcement, usually arranged symmetrically in the inner belt structure parallel to the length of the belt.

3.14 Transverse seam. The junction of material edging across the length of the finished belt.

Table I - Classification of defects

Defect	Major	Minor	Figure number
Indentation, any side	X	-----	1
Indentation, any side	X	-----	2
Indentation, any side	X	-----	3
Indentation, any side	X	-----	4
Indentation, wearing side	X	-----	5
Indentation, top or bottom	X	-----	5
Light spot	X	-----	6
Light spot	-----	X	7
Press lap defect	X	-----	8
Press lap defect	X	-----	9
Flow distortion	X	-----	10
Flash distortion (here shown with flash in place but will appear usually for inspection with flash removed)	X	-----	11
Flash untrimmed (not acceptable but can be removed)	X	-----	12
Misplaced brand with twist (twist revealed by bending belt as shown)	X	-----	13
Misplaced brand without twist	-----	X	14
Protrusion, size shown, any side	X	-----	15
Protrusion, 1/2 size shown wearing side	X	-----	15
Protrusion, 1/2 size shown top or bottom	-----	X	15
Blister, any side	X	-----	16
Foreign material, wearing side	X	-----	17
Foreign material, top or bottom	-----	X	17
Foreign material, internal	X	-----	18
Hole, in cogging area	X	-----	19
Porosity, not in cogging area	X	-----	20
Open longitudinal seam, any side	X	-----	21
Open longitudinal seam, wearing side	X	-----	22
Open longitudinal seam, top or bottom	-----	X	22
Loose longitudinal seam, any side	X	-----	23
Open transverse seam	X	-----	24
Open transverse seam, wearing side	X	-----	25
Open transverse seam, only top or bottom and no larger than shown	-----	X	25
Loose transverse seam	X	-----	26
Loose transverse seam, wearing side	X	-----	27
Loose transverse seam, top or bottom	-----	X	27
Torn cover	X	-----	28
Torn cover, wearing side	X	-----	29
Torn cover, top or bottom	-----	X	29
Trim cut	X	-----	30
Trim cut	-----	X	31
Scratch, not through a ply	-----	X	32

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

Table I - Classification of defects, (Cont'd)

Defect	Major	Minor	Figure number
End scrap, material missing	X	-----	33
Wrinkle in cover, tension member not distorted	-----	X	34
Cogging cut-outs skipped, more than one cog	X	-----	35
Cogging cut-out overlap	-----	X	36
Cogging missed, more than one cog	X	-----	37
Wavy wire in steel cable belt	X	-----	38

Defects not shown on photographs

Defect	Major	Minor	
Cogging cut-out skipped	X	-----	
Cogging missed, one cog	-----	X	
Hole, not in cogging area	X	-----	
Marking, with wrong information	X	-----	
Marking, missing or illecible	X	-----	
Matching faulty	X	-----	
Porosity in cogging area	X	-----	
Scratch through ply	X	-----	
Wrinkle in cover, tension member distorted	X	-----	

4. GENERAL REQUIREMENTS

4.1 Sampling. All belts shall be subjected to one or more sampling procedures in accordance with MIL-STD-105, as specified by the purchaser, except where the material specification includes a sampling plan. Acceptance or rejection of the lot, sublot, or batch shall be on the same basis.

Custodians:
Army - ME
Navy - SH
Air Force - 82

Review activities:
Army - ME, MI
Navy - SH, YD
Air Force - 82

User activities:
Army - AT, EL
Navy - AS

Preparing activity:
Navy - SH
(Project 3030-0051)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 1. Indentation on bottom of belt - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 2. Indentation on top of belt - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

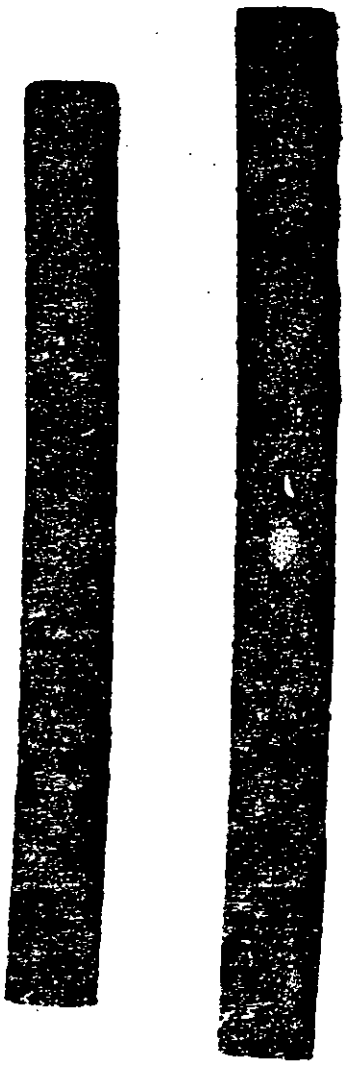


Figure 3. Indentation on top of belt - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 4. Indentation on top of belt - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 3. Indentation on top of bolt - major

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 6. Light spot - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

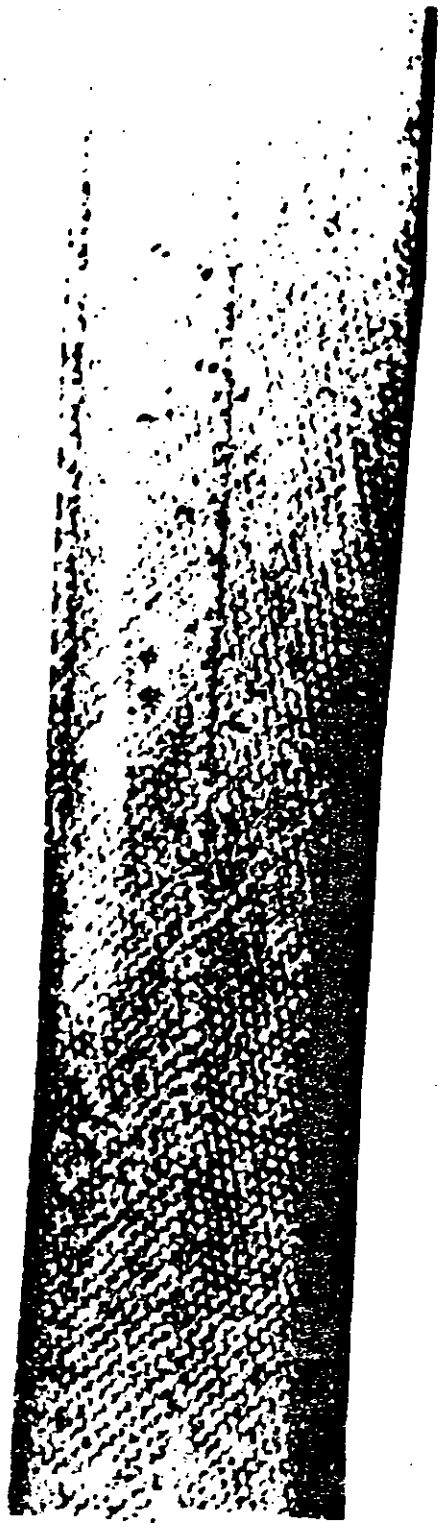


Figure 7. Light spot - minor (upper limit for minor, any larger becomes major).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 8. Press lap defect - major

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 9. Press lap defect - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 10. Flow distortion - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Fig. 11. Flash distortion - major

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

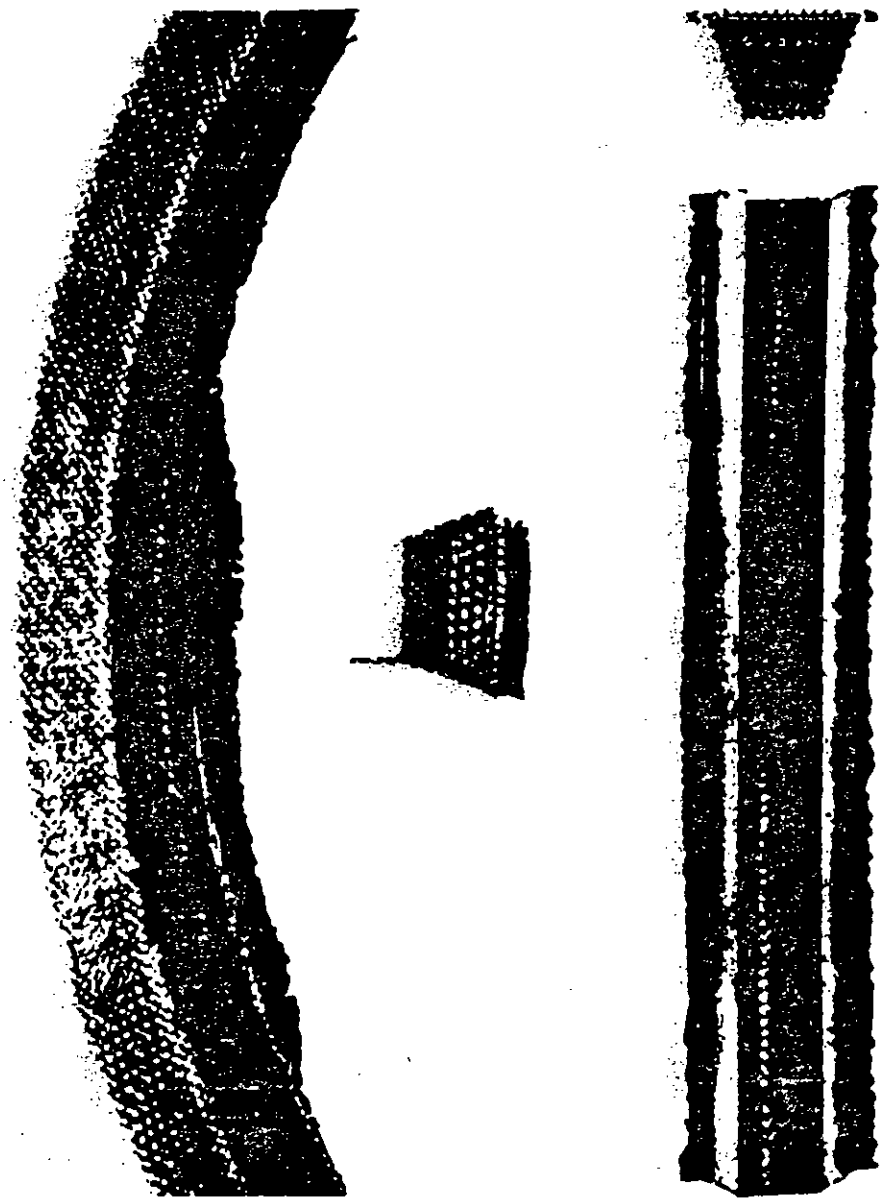


Figure 12. Flash untrimmed - major

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

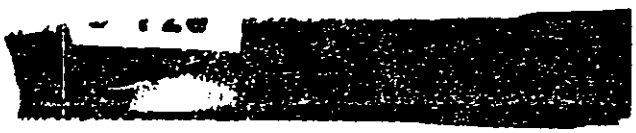


Figure 11. Misplaced strand with twist - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

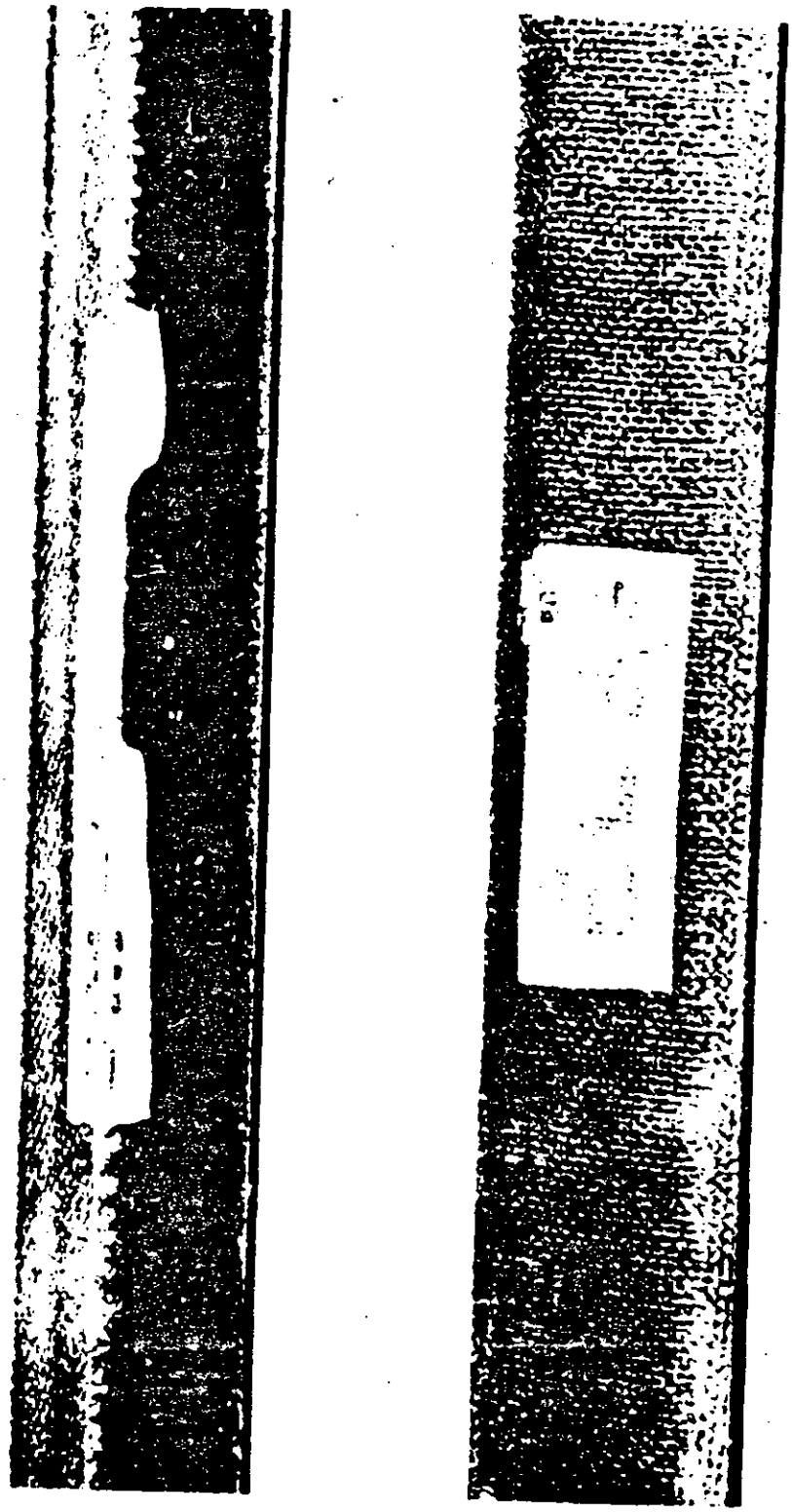


Figure 14. Misplaced brand without twist - minor. (Left belt has misplaced brand. Right belt is a correct one for comparison).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 19. Protrusion - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 16. Blister, any side - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 17. Foreign material, top of belt - minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

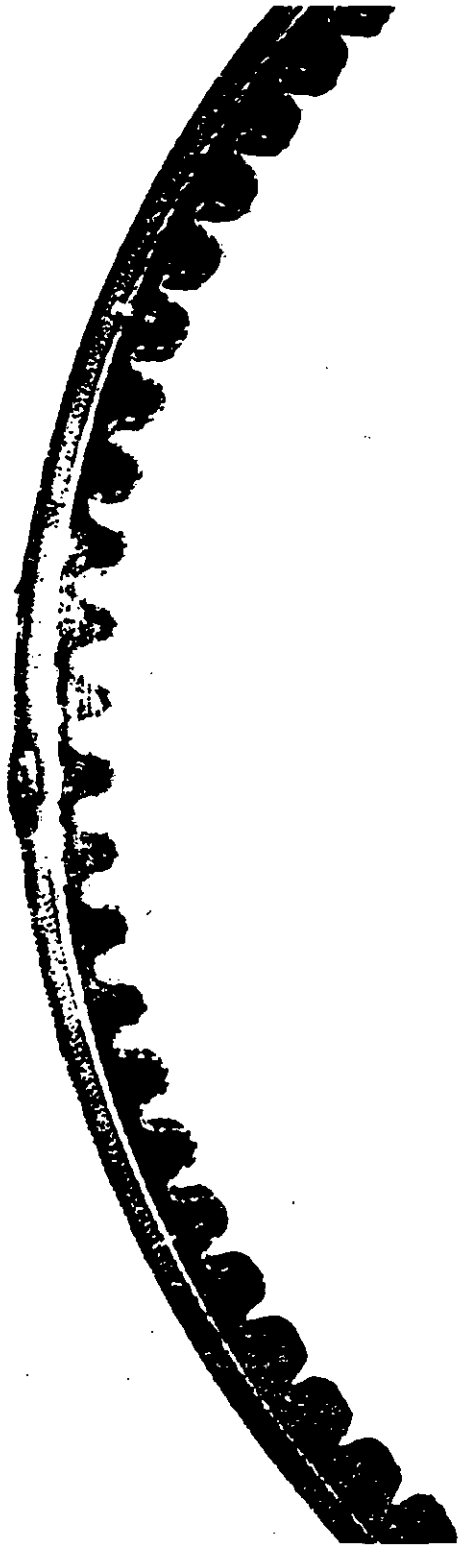


Figure 18. Foreign material, internal - major

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

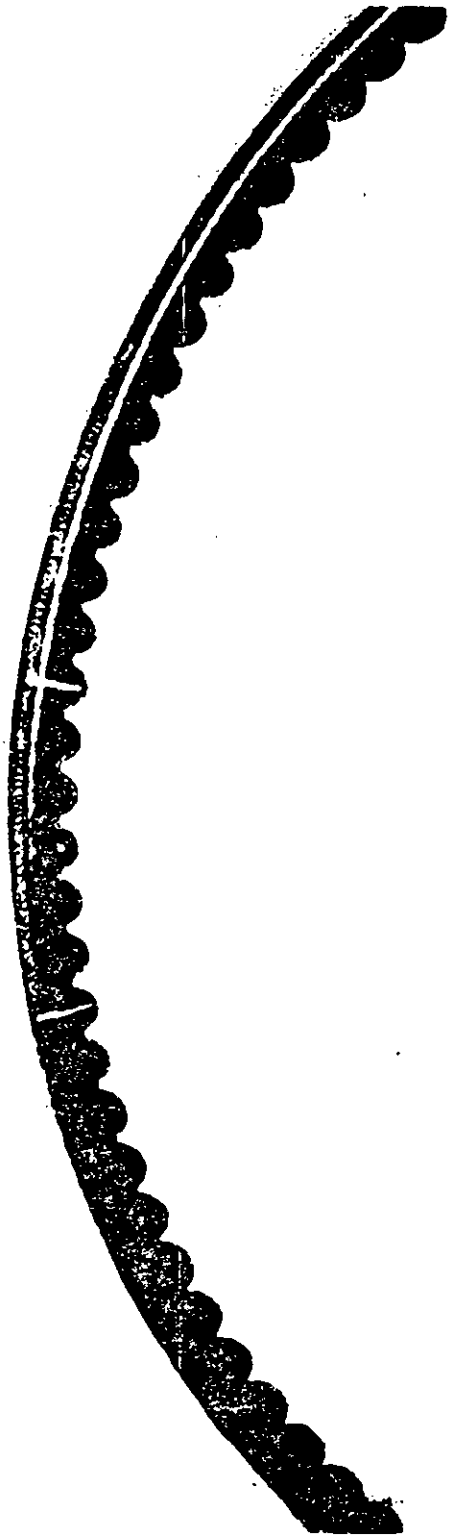


Figure 19. Hole in cogging area - major

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

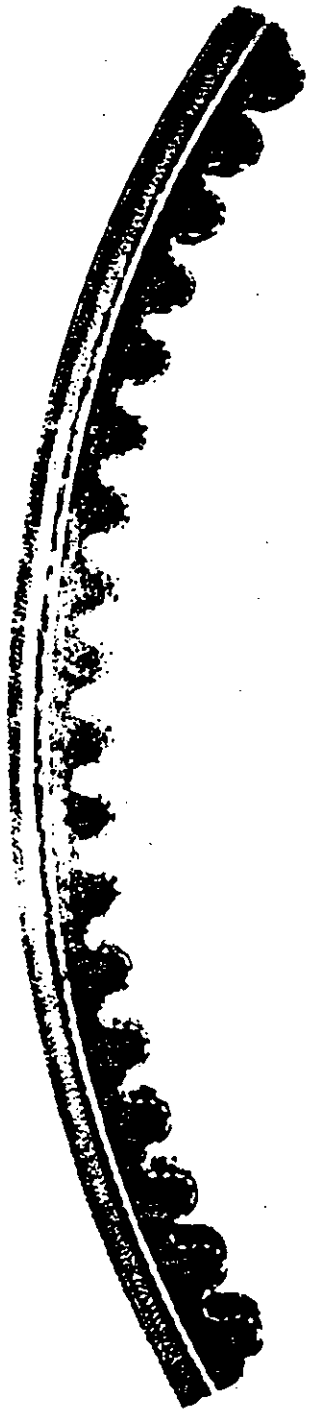


Figure 20. Porosity, between cord layer and upper part of belt - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 21. Open longitudinal seam - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 22. Open longitudinal seam, bottom of belt - minor; if on wearing side - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 23. Loose longitudinal seam - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 24. Open transverse seam - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z .
Check the source to verify that this is the current version before use.



Figure 25. Open transverse seam - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 26. Loose transverse seam - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 27. Loose transverse seam, wearing side - major; top or bottom - minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 28. Torn cover - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 29. Torn cover, bottom of belt - minor; if on wearing side - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

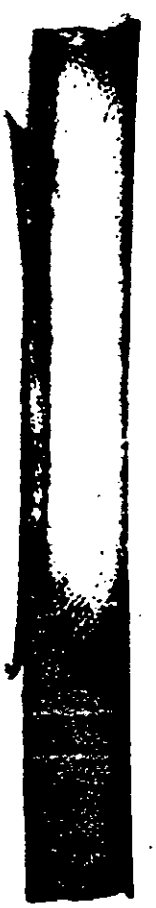


Figure 30. Trim cut in cover - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

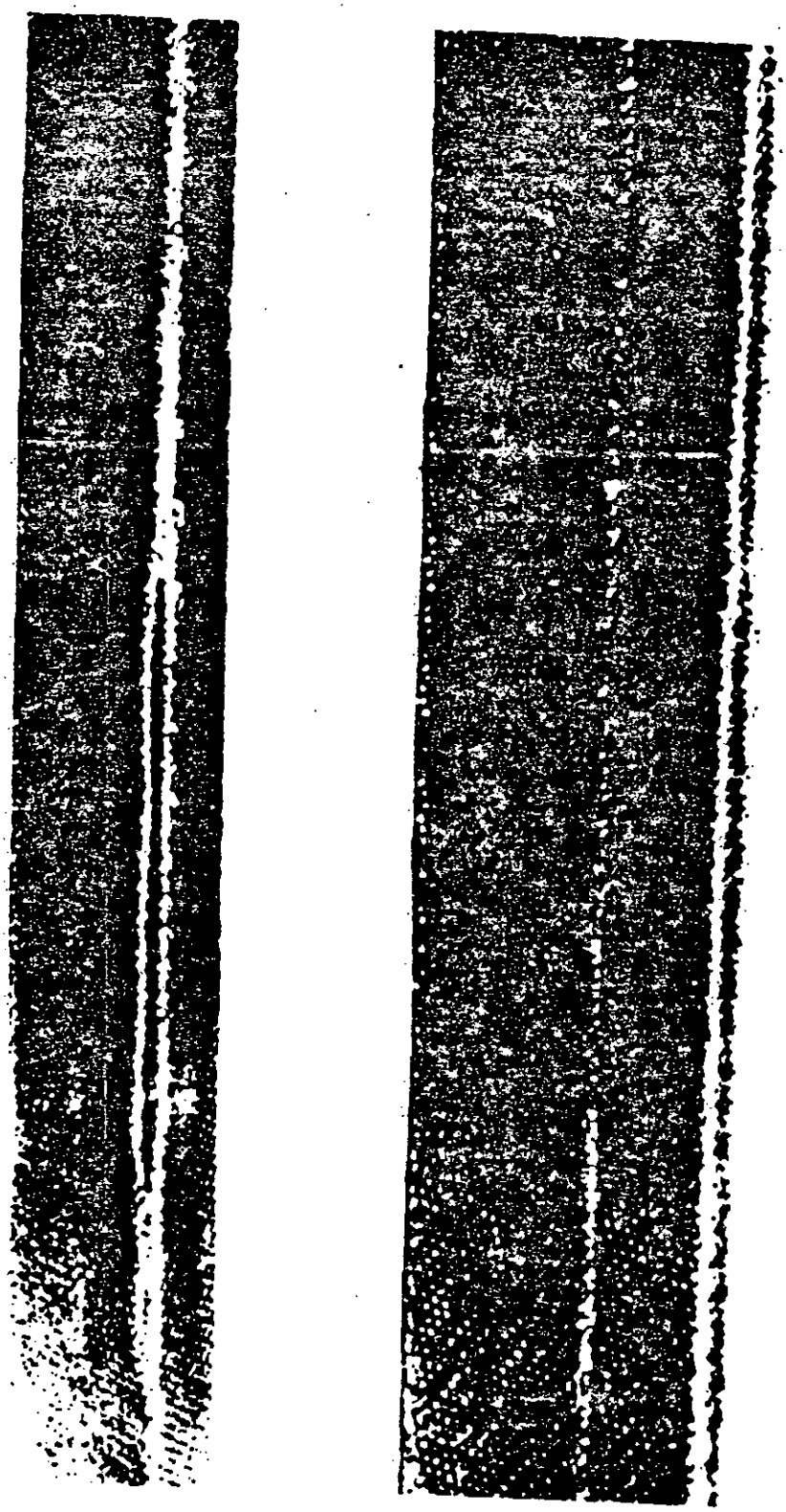


Figure 31. Trim cut in cover - minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

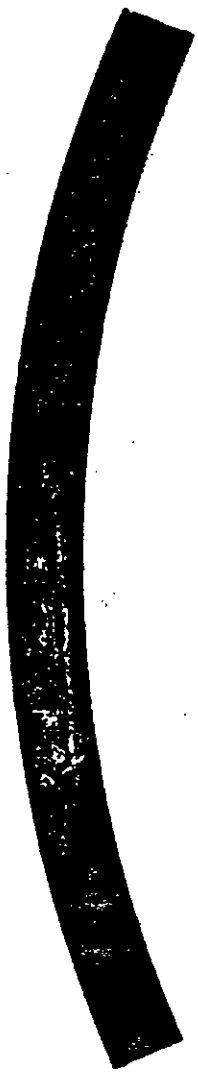


Figure 32. Scratch, not through a ply - minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

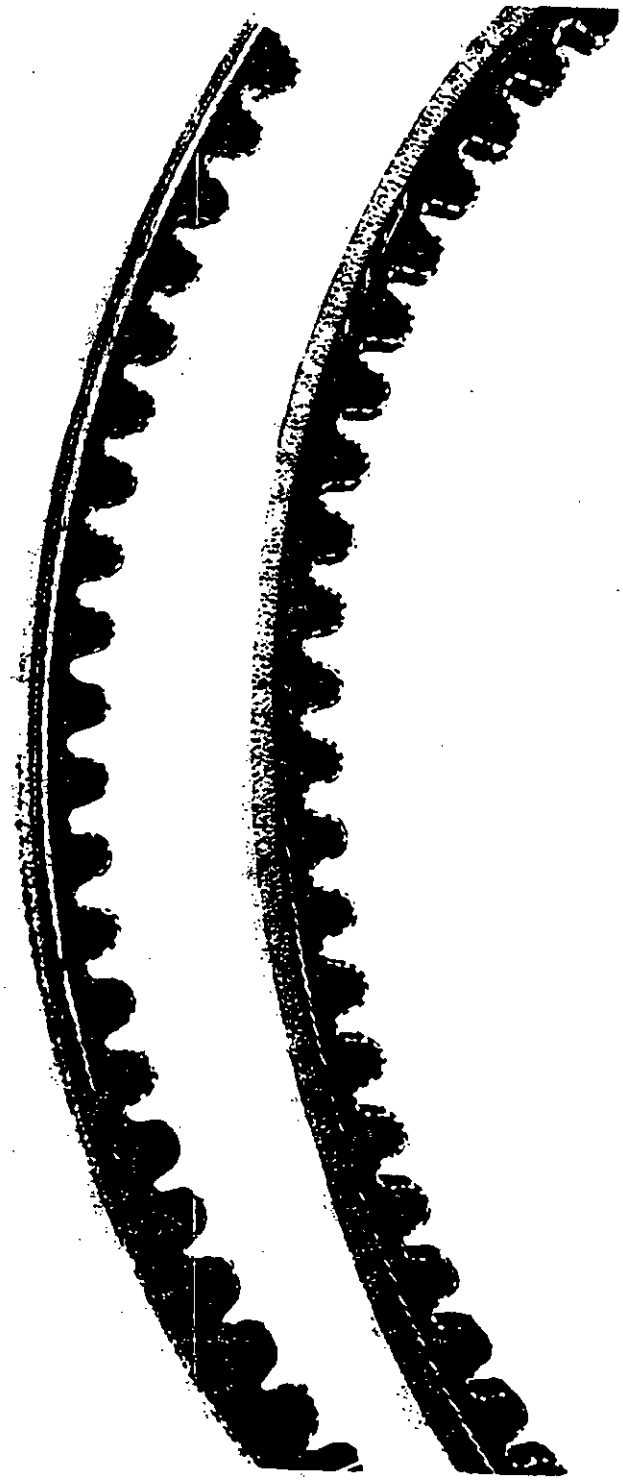


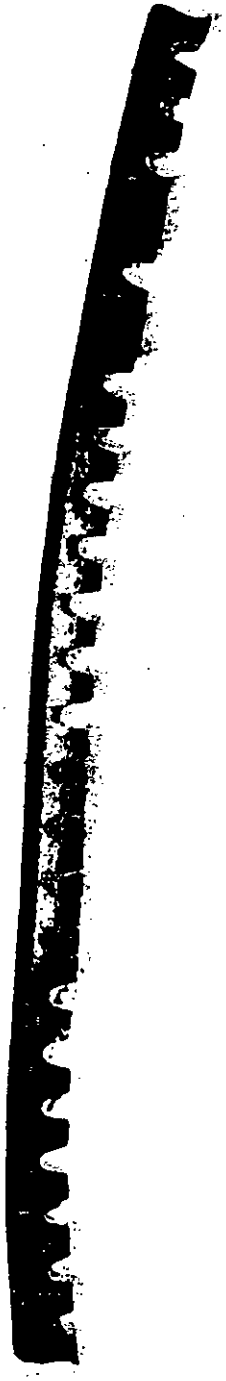
Figure 33. End scrap (Left belt has material missing due to being cut too near the end of the mandrel. Right belt is a correct one for comparison.) - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



Figure 34. Wrinkle in cover - minor.

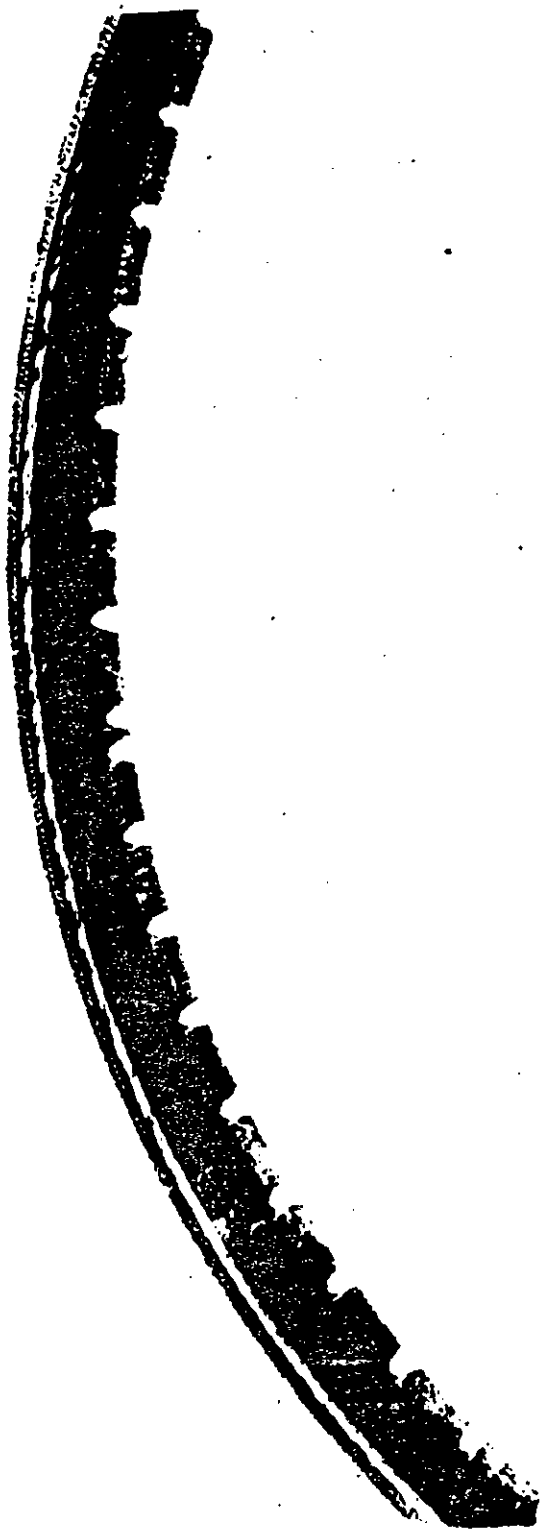
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



SH9373

Figure 35. Cogging cut-outs skipped - major.

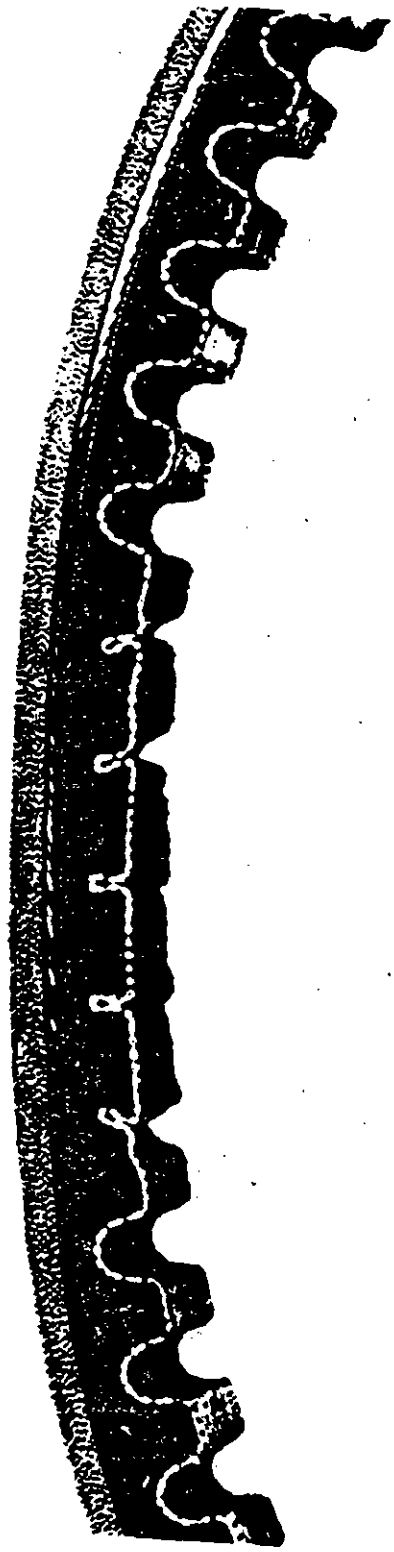
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



SH9374

Figure 36. Cogging cut-out overlap - minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.



SH3375

Figure 37. Cogging missed - major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.

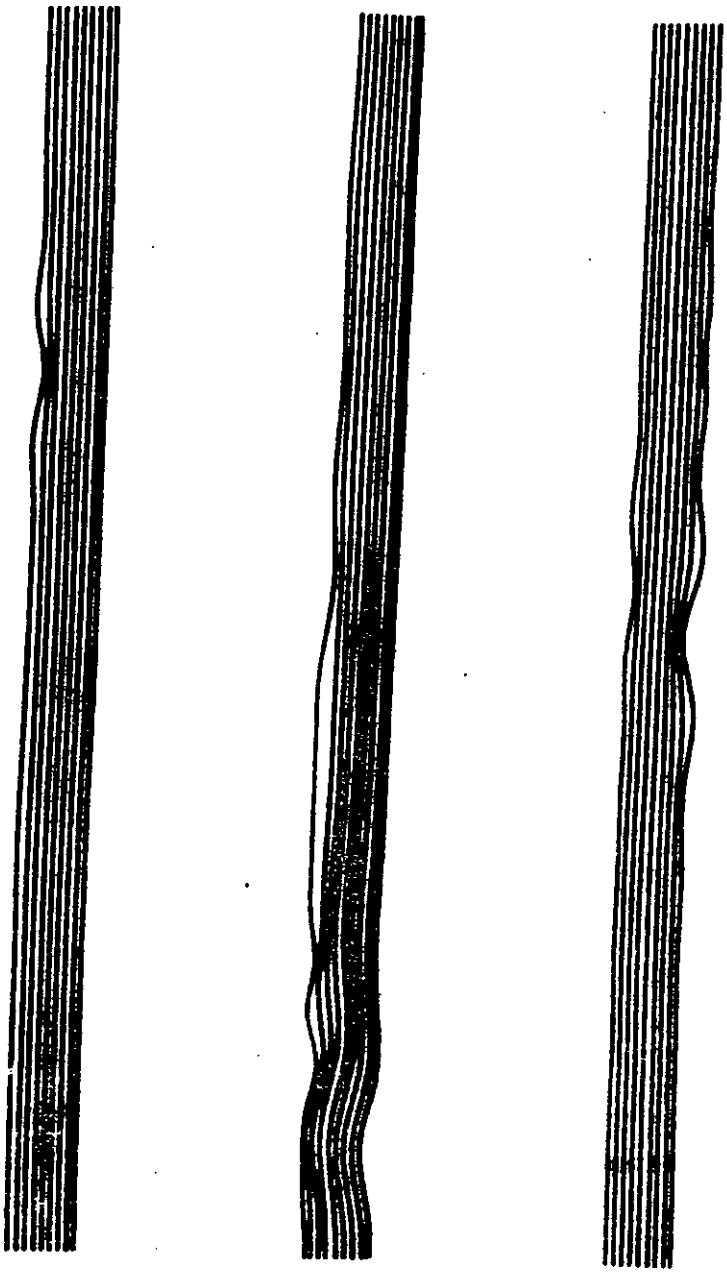


Figure 38. Wavy wire in steel cable belt (X-ray view) - major.

U.S. GOVERNMENT PRINTING OFFICE: 1969-393-088/5245

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T21:54Z
Check the source to verify that this is the current version before use.